

Understanding/Analyzing Digital Audio Channel Status Bits



Glen Kropuenske
SENCORE, Inc.

Application Engineer

1.800.736.2673 or 1.605.339.0100

<mailto:ae@sencore.com>

<http://www.sencore.com>

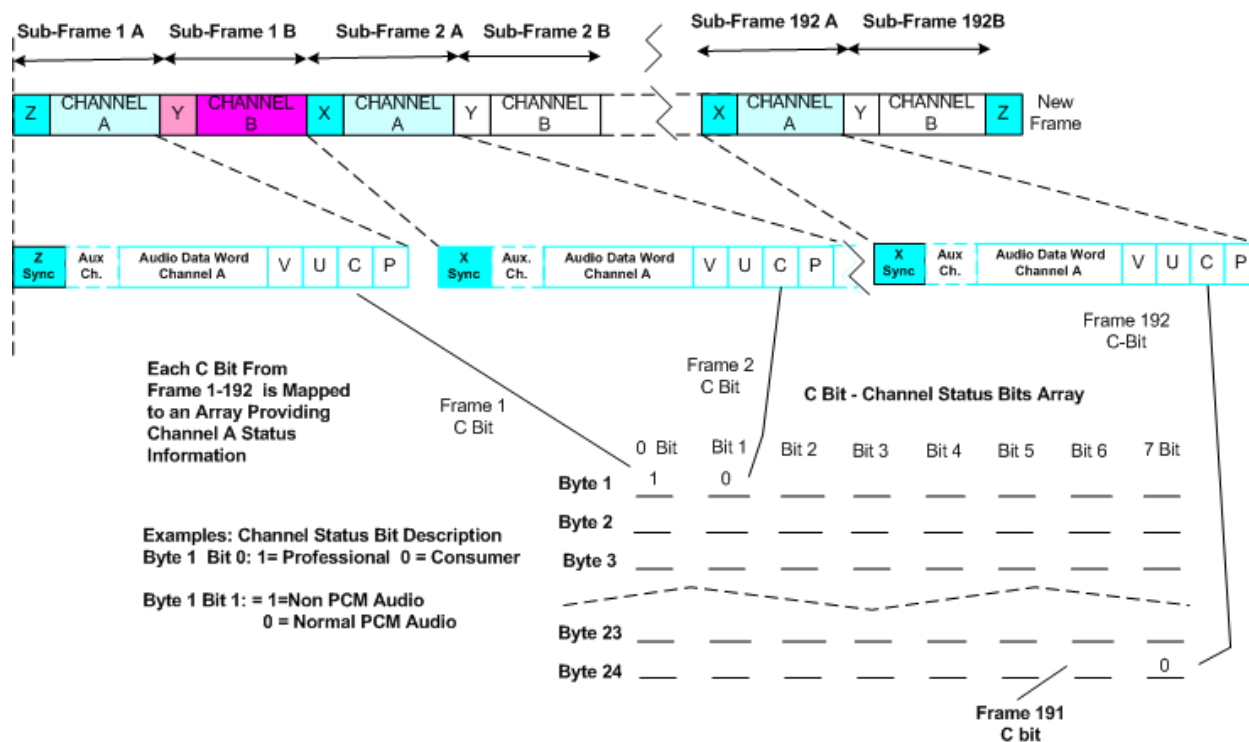
A digital audio receiver extracts frame by frame the channel status (C) bit of each digital audio sub-frame. The C bits are put into a memory bit array for deciphering information about the digital audio channel. Channel status or C bit information can influence how a particular receiver or digital audio device processes or manages the digital audio signal.

This technical article describes what information is carried in the channel status bits, and explains how it can be extracted and interpreted for both the consumer and professional AES formats. It further explains how the Sencore DA795 Digital Audio Analyzer can be used to read and analyze the channel status bits of a digital audio signal.

Channel Status Bit (C Bit) Interpretation

Each of the digital audio sub-frames contains a channel status C bit. A digital audio block contains 192 channel sub-frames for both channel A and channel B so there are 192 channel status bits per channel. The bits are assigned to carry status information regarding the channel's digital audio information.

“C” or Channel Status Bit Mapping/Interpretation



Channel Status bits are assembled into 24 bytes with 8 bit words as illustrated.

The Channel Status Bits are clocked out of each sub-frame into a memory array for interpretation. Understanding the memory array and the assembly of bits from subsequent sub-frames holds the clue for understanding how to interpret the channel status information. The array can be viewed as a table of information containing 8 bit bytes.

Starting with sub-frame 1, the first C bit is read as the 0 bit of byte 1. The second sub-frame's status bit is read into bit 1 of byte 1 and so on. In this manner, each C bit is read into an array, ending with the C bit from the 192nd sub-frame being mapped into bit 7 of byte 24.

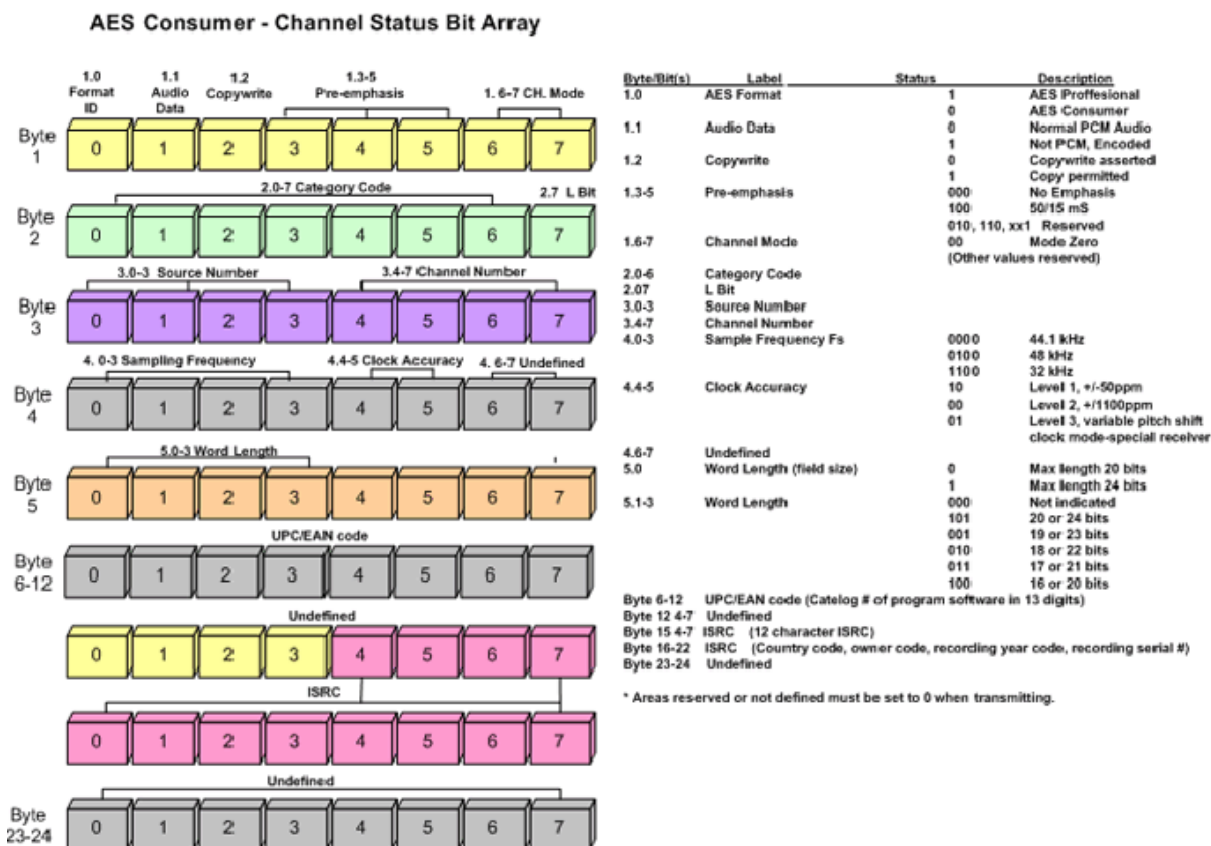
With each C bit mapped into the array, the C bits can be grouped for interpretation. In some instances a single bit indicates something concerning the channel's audio. In other instances, bits are grouped together to provide information.

The channel status bits are grouped differently and contain different meaning depending on whether the digital audio stream is identified as a professional AES digital audio stream or a consumer AES digital audio stream. The first bit extracted from the digital audio block in sub-frame A, bit 0 identifies the audio as professional or consumer AES digital audio. Each additional channel status bit is covered briefly in the following sections.

Consumer Format - Channel Status Bits

Once the AES digital audio is identified as a consumer format, the remaining channel status bits can be arranged into the bit array for interpretation. The information contained by the channel status bits can be described by placing the bits into a table containing each of the 24 bytes of data as shown in the nearby figure. The table has the same row and bit structure as the memory array shown earlier.

Once the digital audio equipment reads the channel status data, it configures itself as instructed by the channel status bits. If improperly instructed or improperly interpreted, the audio equipment does not work properly. This becomes important when things don't work as expected, that you have a way to inspect the channel status bits.



Summary of the Channel Status Bit Array of an AES Consumer Digital Audio signal

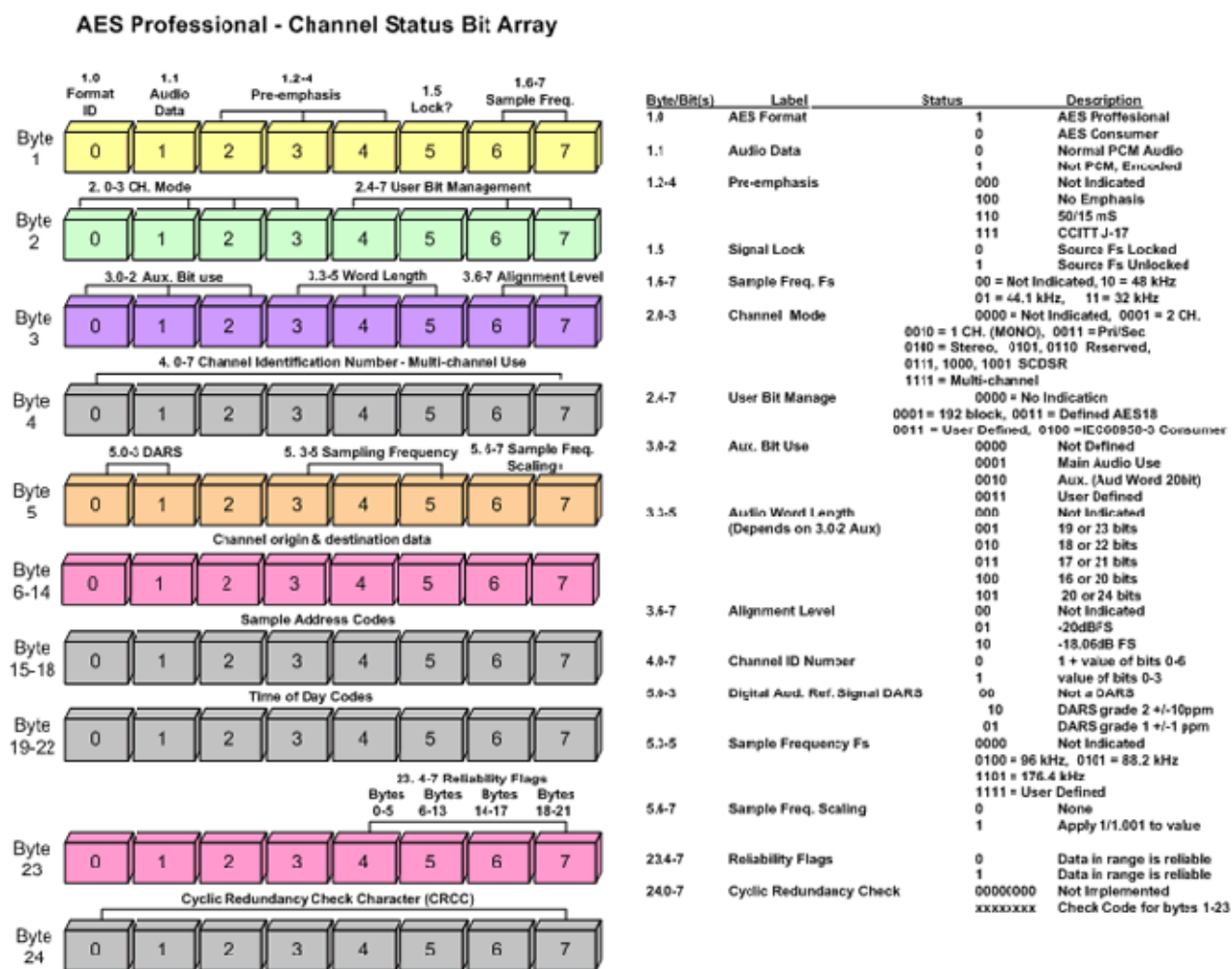
The channel status bits include information regarding whether the audio data is a standard PCM audio or if it has been modified by a compression scheme. The second bit of byte 1 (1.1) indicates if the audio is a linear PCM (logic 0) or if the audio has been compressed (MPEG, DTS, Dolby), which are non-linear PCM formats (logic 1). If this bit is set to "1", a normal PCM audio receiver mutes the audio output.

A copyright bit (1.2) indicates whether the digital audio can be copied (logic 1) or if the audio is copy protected or prohibited (logic 0). The emphasis field (1.3-5) indicates whether the receiver should apply de-emphasis after the audio is converted to analog. The only emphasis supported by the consumer format is a CD or 50/15 mS type. The field may also indicate that no de-emphasis should be applied by the D/A when processing the digital signal.

Other channel status bits indicate the category code, L bit, digital audio word length, sample frequency, source and channel number, clock accuracy as indicated by the consumer channel status bit array. The nearby figure lists these additional information fields and describes the bit structures in those that are commonly used.

Professional Format – Channel Status Bits

The AES professional channel status bit array indicates the use of the channel status bits. The arrangement of channel status bit information differs from the consumer table, although much of the same information is included. Please see the nearby figure for a description of the fields within the professional channel status bits, along with values within each field and a description.



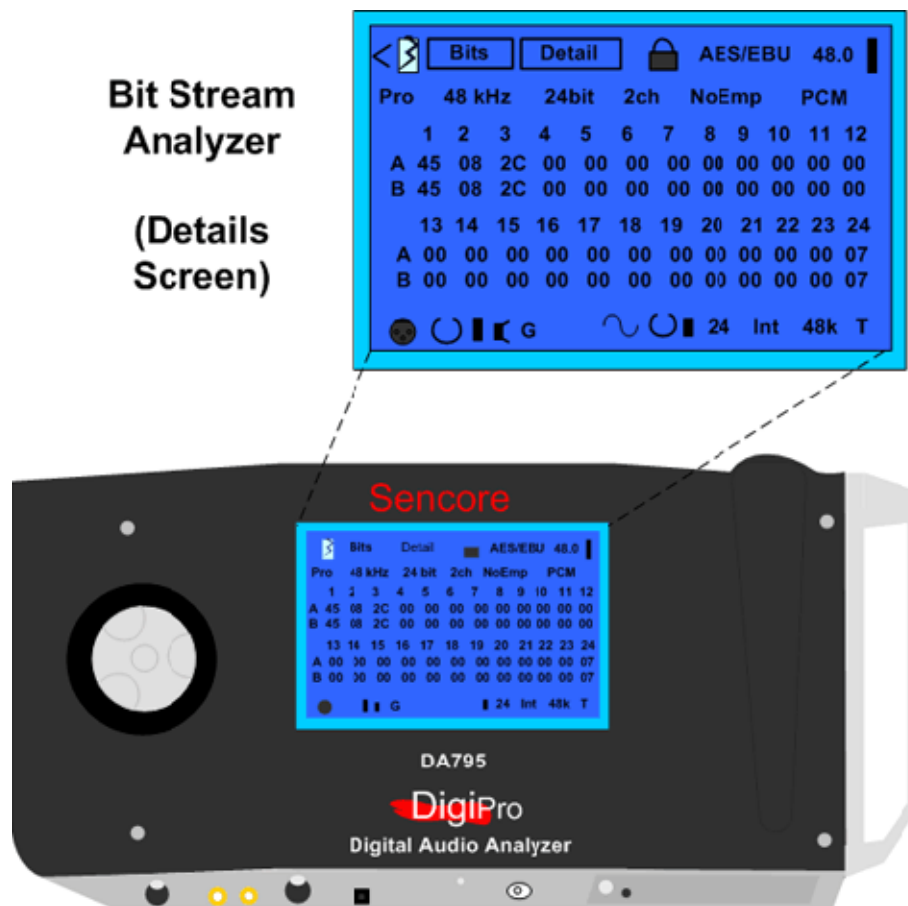
Summary of the Channel Status Bit Array of an AES Professional Digital Audio signal

Reading Channel Status Information with the Sencore DA795

The Sencore DA795's Bit Stream Analyzer function analyzes a digital audio input and displays a thorough analysis of the digital audio signal. Within the "Detail" screen of this function the DA795 reads and displays the digital audio's channel status bits for both consumer and professional AES formats.

The “Detail” screen contains two sections which provide information regarding the channel status bits of the digital audio input selected by the analyzer. The input to the analyzer may be through a balanced AES/EBU input or S/PDIF input.

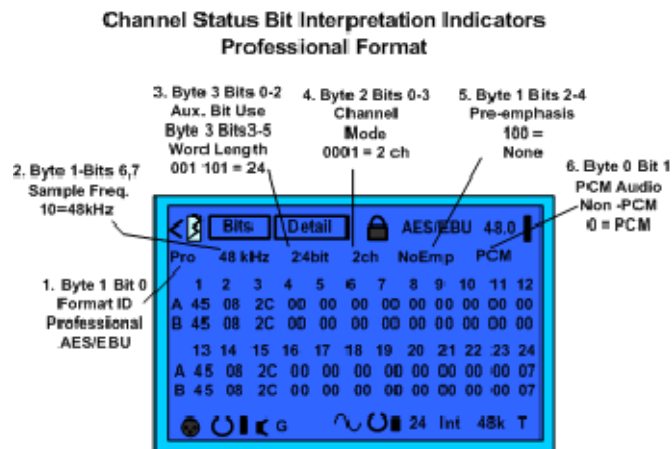
For ease in interpretation, several channel status bits are read by the DA795 and displayed as interpreted data in the second line of the "Detail" screen. A complete analysis of all 192 bits of the channel status bits are indicated by 24 hex values for both the channel A and channel B sub-frames. These two indicators are described in the following sections.



The Sencore DA795 Digital Audio Analyzer provides details of the channel status bit information.

The second line of the DA795 contains readouts which are interpreted information from the digital audio's channel status bits. The readout fields indicated by the DA795 for a professional digital audio AES format are listed below and shown in the nearby figure.

1. Professional or Consumer Digital Audio Format (Byte 1.0)
2. Sample Frequency (Byte 1.6-7)
3. Audio Word Length (Byte 3.0-2 Aux, Byte 3.3-5)
4. Channel Mode (Byte 2 0-3)
5. Pre-emphasis (Byte 1.2-4)
6. PCM Audio or Encoded Audio - nonPCM (Byte 0 bit 1)

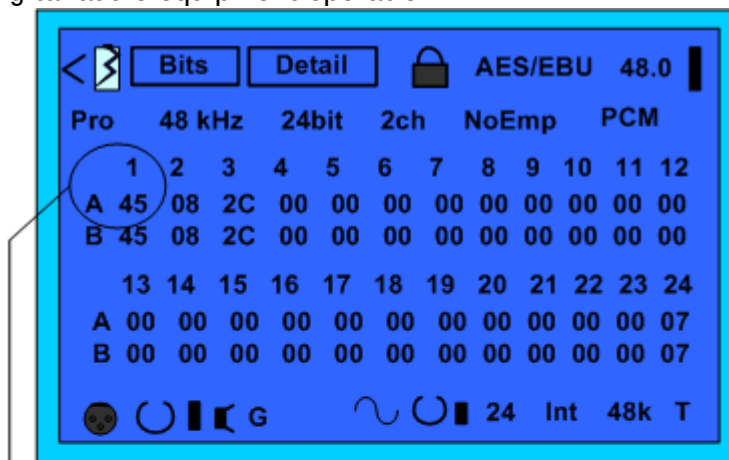


The second line of the “Detail” screen includes readouts of several of the channel status bit fields.

Determining the Channel Status Bit Values

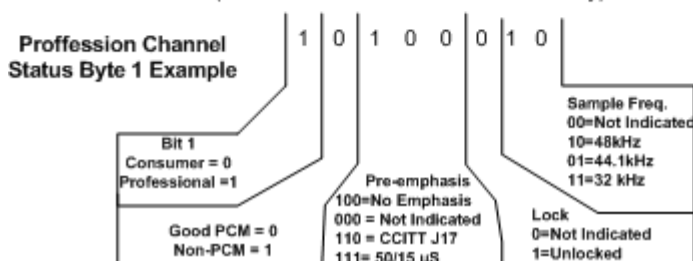
The DA795 can be used to determine the value of all the channel status bits for comparison to the channel status bit table. The values can be compared to known good values for fast diagnoses, or can be used to determine the cause of improper digital audio equipment operation.

The DA795's Bit Stream Analyzer function includes a "Detail" screen which shows a hex value for all 24 frames of the channel A and B sub-frames. These hex values are shown in the middle part of the "Detail" screen in rows for channel A and channel B under numeric headings of 1 to 24. When converted to binary, the value of each bit in the channel status bit array is indicated.



CHANNEL A - Byte 1 of Channel Status Bits:
Hex = 45 - Converted to Binary = 01000101
(Reversed to match channel status array)

Example of the 1st byte in the channel status screen changed from hex to binary with an interpretation of the bits according to the channel status table.



To determine the values, the screen is divided into 24 fields to match the number of bytes in the channel status bit array. The channel status bits from both the channel A and channel B sub-frames are read and indicted as hex values under the byte heading. The first step is to convert the hex value to a binary equivalent. Fortunately, there are handy conversion charts for this. Second, turn the binary bits around so the LSB becomes the MSB or moves to the left side of the page. The binary value orientation then matches the bytes as listed in the channel status table.

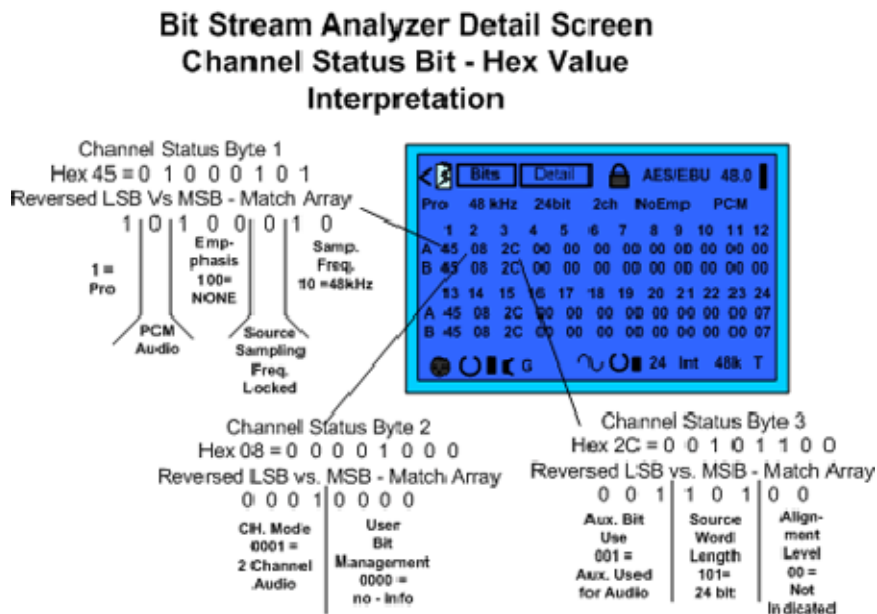
For example, in the nearby figure, byte 1 for channel A is defined as hex value "45." The hex value "45" converted to binary results in a value of 01000101. While this might seem difficult, a very simple chart can be used. Click on this link for an easy HEX to BINARY Conversion Chart:

<http://www.sencore.com/newsletter/Nov05/BinaryConversionTable.htm>.

Once the binary value is determined from the chart, reverse the order of the digital byte or change the least significant bit (LSB) to the most significant bit (MSB). This is done to match the order of the channel status bit array which was determined from left to right on the page. Now the bit orientation matches the order of the channel status bit table and can be interpreted.

NOTE: Be sure to apply the binary value to the correct byte or row in the channel status bit table for accurate results. Some technical information on this same subject start with byte 0 rather than byte 1 for the 1st row as is used in this technical document.

The 24 channel status bytes for both channel A and B are indicated in the “Details” measurement screen.



The nearby figure shows the second byte of the channel status bits to be a hex value of “08.” Notice that this is the same value for channel A and channel B. For this reason, many receivers only read one of the channels status bits. Hex value 08 is equivalent to binary value 00001000. When the binary bits are reversed to indicate the LSB value on the left to match the channel status bit array tables the value is 00010000. The value indicates 2 Channel audio and no user bit information.

The Future is Digital

The Sencore DA795 Digital Audio Analyzer is ready to help you with your digital audio challenges. For more information on the DA795 see <http://www.sencore.com> or call 1.800.736.2673.

Learn more – DA795 DigiPro Digital Audio Analyzer: <http://www.sencore.com/products/da795.htm>

<http://www.sencore.com>

<mailto:sales@sencore.com>

1.800.736.2673 or
1.605.339.0100